Template Matching, Not Time Learning

A Diagnostic for Self-Supervised Light-Curve Encoders

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The question

Ground-based surveys (ZTF, soon Rubin LSST) sample stellar light curves on a wildly irregular time grid. Self-supervised encoders are the standard way to learn from $\sim 10^9$ unlabeled sources — and a popular test of whether they have "learned time" is **period regression**: fit a probe from the embedding to $\log_{10} P$ and report R^2.

But high R^2 is ambiguous. An encoder gets period "for free" if it merely identifies the *class* (RR Lyrae $\sim$0.5 d, Mira $\sim$100 d) and the probe predicts the class-mean period — *without ever reading period off the individual time series.*

The diagnostic

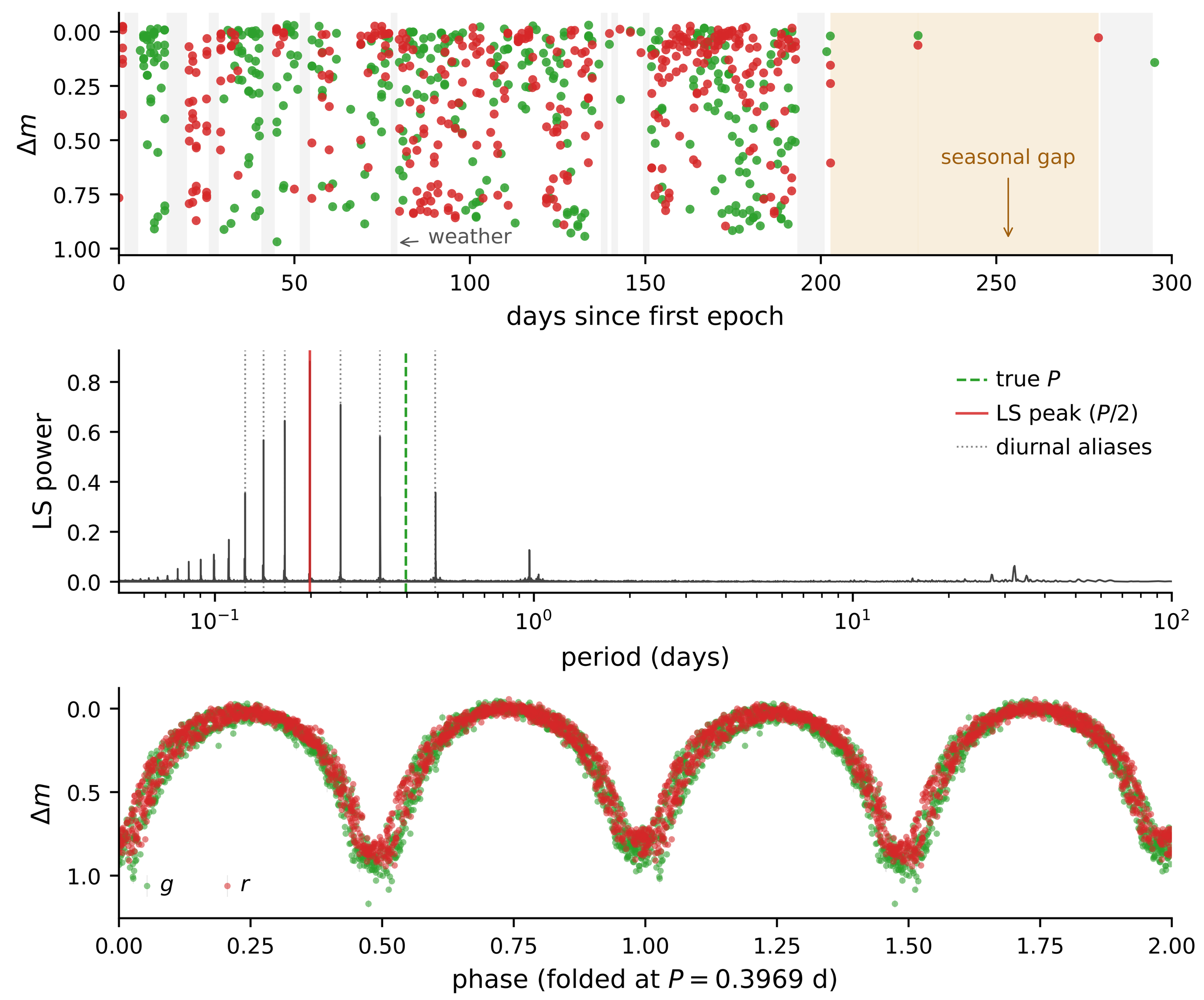
Decompose the variance of the *predicted* $\log_{10} P$ into a between-class and a within-class part:

$$f_{\text{cls}} = \frac{\Sigma_c n_c (\bar{\hat{y}}_c - \bar{\hat{y}})^2}{\Sigma n \cdot \operatorname{Var}(\hat{y})} \qquad \rho_{\text{within}} = \underset{c}{\text{median}}\ \rho_{\text{Spearman}}(y, \hat{y} \mid c)$$

- $f_{\text{cls}} \to 1$: predictions constant within a class $\Rightarrow$ **class-template matching**.
- $\rho_{\text{within}} \to 1$: sources ordered by true period *inside* each class $\Rightarrow$ **genuine time learning**.

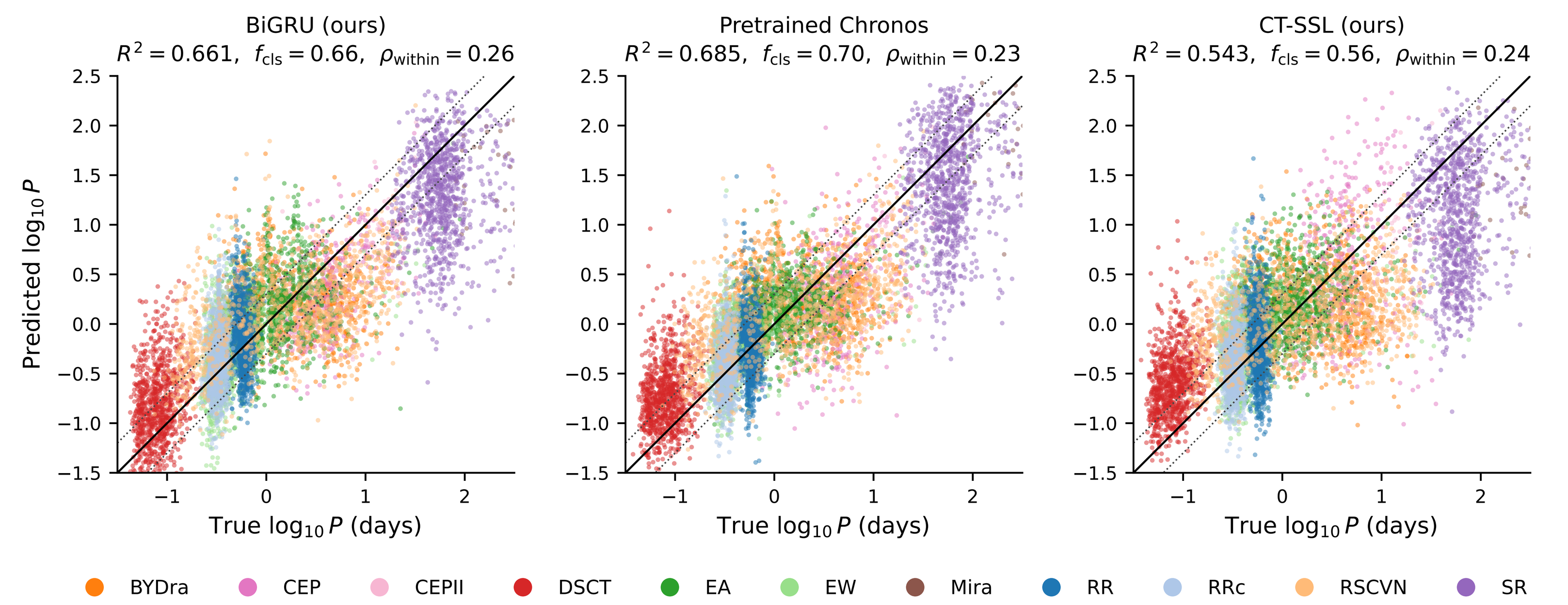
Learned absolute time $\equiv$ high R^2 *and* low f_{cls} *and* $\rho_{\text{within}} \to 1$.

What the encoder sees



Only (flux, $\log_{10}\sigma$, $\log_{10}\Delta t$) per epoch — never the period, phase, or periodogram.

Negative result: every encoder template-matches



Across **eight encoders** — pretrained Chronos-T5 (46M) and MOMENT (110M), a 4.4M BiGRU, our 4.8M continuous-time SSL transformer — the diagnostic returns the *same* answer:

60–70% of period R^2 is between-class.
Within-class $\rho \le 0.26$ (linear) for *every* method.
Each class forms a near-vertical stripe at its typical period.

Robustness: nonlinear probe + 95% CIs

Is period information merely *nonlinearly* encoded? We re-ran the diagnostic with a nonlinear MLP probe and bootstrap 95% CIs over 5 splits.

	R^2	ρ_{within}
BiGRU ridge	0.66	0.26 [0.22,0.29]
BiGRU MLP	**0.77**	**0.31** [0.28,0.34]
Chronos ridge	0.69	0.23 [0.20,0.26]
Chronos MLP	0.72	0.27 [0.26,0.32]

The MLP raises R^2 by *sharpening class templates*, but ρ_{within} never exceeds 0.31. **The missing signal is in the encoders, not the probe.**

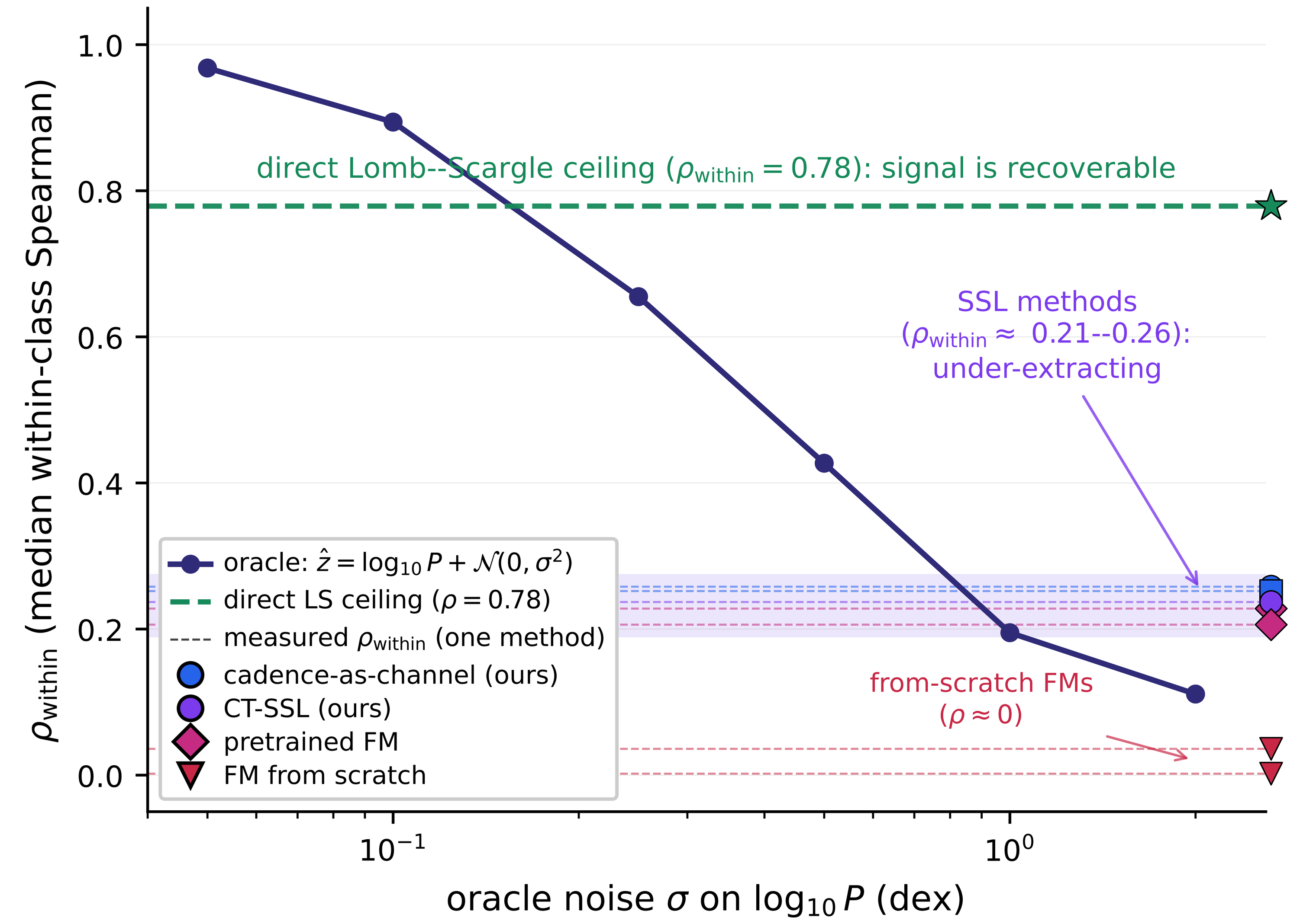
Architectural-fairness control

Retrain Chronos-T5 *from scratch* at matched scale (4.9M) on the same 400K-source ZTF pool:

	Chronos-T5	pretrained	from scratch
class. bal-acc		0.65	0.32
period R^2		0.69	0.19
ρ_{within}		0.23	0.03

Cross-domain pretraining buys **class templates, not time**: even *with* it, $\rho_{\text{within}} = 0.23$ — below our pretraining-free BiGRU (0.26).

The signal *is* recoverable



A direct Lomb–Scargle period extractor on the *same* light curves reaches $\rho_{\text{within}} = 0.78$. The SSL encoders capture only $\sim$1/3 of recoverable within-class ordering — they are **under-extracting**, not saturating a label-noise floor.

Takeaway for the field

1. Period R^2 alone **overstates** learned time encoding.
2. Report f_{cls} and ρ_{within} alongside it; treat ρ_{within} as the **metric of merit**.
3. No tested SSL encoder has learned absolute time on irregular ZTF photometry — it remains an **open problem**.

Code, data & paper

`github.com/emmachickles/period-diagnostic`

Explore a source live

fold at true P vs. the $P/2$ & daily aliases

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References

[1] Chen et al. 2020, ApJS 249, 18 (ZTF periodic-variable catalog). [2] Bellm et al. 2019, PASP 131 (ZTF). [3] Ansari et al. 2024, TMLR (Chronos). [4] Goswami et al. 2024, ICML (MOMENT). [5] Donoso-Oliva et al. 2023, A&A 670 (ASTROMER). [6] Nie et al. 2023, ICLR (PatchTST).